Possible Problems for Quiz 3 to be taken 1/31/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

(a) Suppose vectors $\{\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4\}$ are linearly independent and belong to R^5 do they span R^5? Justify your answer.

(b) Demonstrate that the set of vectors $[0,1,0,0], [1,0,0,0], [0,0,0,1], [0,0,1,0]$ span R^4. Do so by finding the linear combination of the 4 previous vectors that gives the arbitrary vector $[a, b, c, d]$.

(c) Write down a set of vectors that spans R^3 using only the numbers 1 and 0 as components of the vector.

Problem 2

(a) State the definition of span

(b) State the definition of linear independence

(c) State the definition of a linear transformation

Problem 3

Calculate the linear transformation $\frac{1}{2}\begin{bmatrix} \cos\left(\frac{\pi}{2}\right) & -\sin\left(\frac{\pi}{2}\right) \\ \sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right) \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and sketch the vector before and after the transformation. Describe what happened to the vector in words.

Problem 4

Show that the transformation $T(\vec{x}) = \begin{bmatrix} 2x + 3y \\ 4x - y \end{bmatrix}$ is linear.

Problem 5

Show that the transformation $T(\vec{x}) = \begin{bmatrix} 2x + 1 \\ 2y + 1 \end{bmatrix}$ is not linear

Problem 6

Show that if $T(x + y) = T(x) + T(y)\ and\ T(cx) = cT(x)$ (that is, T is linear) then

$$T(c_1x + c_2y) = c_1T(x) + c_2T(y)$$